TOPIK 2C

Optimasi Query Routing pada Arsitektur Read Replica PostgreSQL: Perbandingan 5 Strategi

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1. Deskripsi

Lima strategi routing (Round-Robin, Load-Based, Latency-Based, Weighted Round-Robin, Least-Connections) pada PostgreSQL 1 primary + 3 replica heterogen. Evaluasi pada variasi query complexity dan read/write ratio.

2. Infrastruktur

- PostgreSQL 16 Docker: 1 primary + 3 replica (async). Heterogeneous: 2/1/0.5 CPU. Total ~1.5 GB RAM. Data: 500K rows.
- Custom Query Router Proxy (Python ~200 LOC), 5 strategi pluggable, health check 5 detik, EMA 0.3.

3. Langkah

Pekan 1: PG cluster + replication + data. Pekan 2: Router + 5 strategies. Pekan 3: Integrasi. Pekan 4: Eksekusi + report.

4. Skenario Uji Coba

4.1 Variabel Tetap

- PG 16.x, 3 replica heterogen, 500K rows, async, pool 20, health check 5s, EMA 0.3, 50 concurrent, seed tetap, warm-up 1000 queries.

4.2 Variabel Manipulasi

Variabel	Level
Routing Strategy	Round-Robin, Load-Based, Latency-Based, Weighted-RR, Least-Conn
Query Complexity	Simple (PK lookup), Medium (JOIN 2 tabel), Complex (JOIN 3 + aggregasi)
Read/Write Ratio	Read-Heavy (95:5), Balanced (70:30)

Total: 5 strategi × 3 complexity × 2 ratio = 30 × 5 rep = 150 run × 10 menit = ~25 jam (3 hari). Prioritas: Read-Heavy = 15 × 5 = 75 run (~12.5 jam).

4.3 Variabel Respon

Variabel	Satuan	Cara Pengukuran
Read Avg Latency	ms	Rata-rata read query
Read P95 Latency	ms	Persentil ke-95 read
Overall Throughput	qps	Query berhasil per detik
Load Distribution CV	float	CV query count across replicas

Per-Replica CPU	%	Rata-rata CPU per replica
Staleness Rate	%	Read yang return stale data
Router Overhead	ms	Tambahan latency routing

4.4 Template Pengumpulan Data

Tabel 4.4a — Read-Heavy (95:5), Query Complexity: Simple (PK lookup)

Metrik	RoundRobin	Load-Based	Latency-Based	Weighted-RR	Least-Conn
Read Avg (ms)	___±___	___±___	___±___	___±___	___±___
Read P95 (ms)	___±___	___±___	___±___	___±___	___±___
Throughput (qps)	___±___	___±___	___±___	___±___	___±___
Load CV	___±___	___±___	___±___	___±___	___±___
Staleness (%)	___±___	___±___	___±___	___±___	___±___
Router Overhead (ms)	___±___	___±___	___±___	___±___	___±___

Tabel 4.4b — Read-Heavy (95:5), Query Complexity: Medium (JOIN 2 tabel)

Metrik	RoundRobin	Load-Based	Latency-Based	Weighted-RR	Least-Conn
Read Avg (ms)	___±___	___±___	___±___	___±___	___±___
Read P95 (ms)	___±___	___±___	___±___	___±___	___±___
Throughput (qps)	___±___	___±___	___±___	___±___	___±___
Load CV	___±___	___±___	___±___	___±___	___±___
Staleness (%)	___±___	___±___	___±___	___±___	___±___
Router Overhead (ms)	___±___	___±___	___±___	___±___	___±___

Tabel 4.4c — Read-Heavy (95:5), Query Complexity: Complex (JOIN 3 + aggregasi)

Metrik	RoundRobin	Load-Based	Latency-Based	Weighted-RR	Least-Conn
Read Avg (ms)	___±___	___±___	___±___	___±___	___±___
Read P95 (ms)	___±___	___±___	___±___	___±___	___±___
Throughput (qps)	___±___	___±___	___±___	___±___	___±___
Load CV	___±___	___±___	___±___	___±___	___±___
Staleness (%)	___±___	___±___	___±___	___±___	___±___
Router Overhead (ms)	___±___	___±___	___±___	___±___	___±___

Tabel 4.4d — Balanced (70:30), Query Complexity: Simple

Metrik	RoundRobin	Load-Based	Latency-Based	Weighted-RR	Least-Conn
Read Avg (ms)	___±___	___±___	___±___	___±___	___±___
Throughput (qps)	___±___	___±___	___±___	___±___	___±___
Load CV	___±___	___±___	___±___	___±___	___±___
Staleness (%)	___±___	___±___	___±___	___±___	___±___

Tabel 4.4e — Balanced (70:30), Query Complexity: Medium

Metrik	RoundRobin	Load-Based	Latency-Based	Weighted-RR	Least-Conn
Read Avg (ms)	___±___	___±___	___±___	___±___	___±___
Throughput (qps)	___±___	___±___	___±___	___±___	___±___
Load CV	___±___	___±___	___±___	___±___	___±___
Staleness (%)	___±___	___±___	___±___	___±___	___±___

Tabel 4.4f — Balanced (70:30), Query Complexity: Complex

Metrik	RoundRobin	Load-Based	Latency-Based	Weighted-RR	Least-Conn
Read Avg (ms)	___±___	___±___	___±___	___±___	___±___
Read P95 (ms)	___±___	___±___	___±___	___±___	___±___
Throughput (qps)	___±___	___±___	___±___	___±___	___±___
Load CV	___±___	___±___	___±___	___±___	___±___
Staleness (%)	___±___	___±___	___±___	___±___	___±___
Router Overhead (ms)	___±___	___±___	___±___	___±___	___±___

4.5 Uji Statistik + Visualisasi

- Kruskal-Wallis + Dunn's: 5 strategi per combination. Two-way ANOVA: strategy × complexity.
- Gini coefficient: fairness distribusi query.
- Radar chart per strategi (5 dimensi). Stacked bar per-replica query distribution. Heatmap: latency, strategy (row) × complexity×ratio (col).